

Untapping the Hidden Potential of Non-Textual Maritime Data Using AI

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BACKGROUND

Industries like maritime and automotive generate massive amounts of audio and video data, only a small fraction of which is relevant for training AI. Engineering data is rarely available in machine-readable form — it's locked in figures, drawings, and diagrams.

APPROACH

AI methods — natural language processing, multimodal embeddings, and semantic search — are applied to extract, organise, and retrieve relevant information from non-textual data and represent it in a structured form.

OUTCOMES

Four planned papers:

- I. Maritime Automatic Speech Recognition and Key-Phrase Extraction.
- II. Semantic Search for Maritime Unstructured Data.
- III. Graph Representations of Maritime Systems.
- IV. Information Extraction from Engineering Graphs and Diagrams.

TIMELINE

Conducted in the Virtual Sea Trial project,
1 Jan 2024 – 31 Dec 2026.

Novia University of Applied Sciences
Åbo Akademi University



Make every minute of maritime audio, video, and 3D searchable like text.

So engineers can test ships *virtually* instead of at sea.

KEY IDEA

Voice + video + 3-D models are first-class data, not byproducts. Treat them like text and a whole search index opens up.

WHAT INDUSTRY GETS

- Less time scrubbing through hours of sea-trial footage.
- Automated tests built from real captured behaviour.
- Engineering drawings that answer questions, not just sit in PDFs.

READ MORE



virtualeatrial.fi
Industry presentation
Vaasa, 2026